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**BIT 2324 GEOGRAPHICAL INFORMATION
SYSTEMS**

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

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LESSON 2

Basics of GIS Software Packages

Learning outcomes

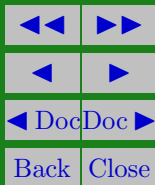
By the end of this topic you should be able to;

- *Explain GIS software*
- *Distinguish between thin and thick client servers*



2.1. Introduction

There are a number of Geographical Information Systems available. They range from high-powered analytical software to visual web applications, and each of those are used for a different purpose. Due to the vast number of GIS available, they have common features. Understanding these basic features will give you confidence with any GIS system that you use. GIS Applications are normally programs with a graphical user interface that can be manipulated using the mouse and keyboard. The application provides menus near the top of the window which, when clicked using the mouse, show a panel of actions. These actions provide a way for you to tell the GIS Application what you want to do. The software encompasses not only the GIS package, but all the software used for databases, drawings, statistics, and





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imaging and the functionality of the software used to manage the GIS determines the type of problems that the GIS may be used to solve. The software used must match the needs and skills of the end user. A common function of GIS Applications is to display map layers. Map layers are stored as files on a disk or as records in a database. Normally each map layer will represent something in the real world. Three groups of GIS will be covered:

- Web-based GIS: ONS and London Profiler
- Geobrowser: Google Earth
- Desktop GIS: ArcGIS



2.2. Desktop GIS

A GIS software allows you to interactively work with spatial data. A desktop GIS is a mapping software that is installed onto and runs on a personal computer. It is software for managing information of any kind according to where it is located. For example, businesses can track customer locations, optimize delivery routes, or decide where to site future businesses using GIS; scientists use GIS to manage sensitive wildlife habitats or track animal movements in an ecosystem; and health care specialists track the spread of infectious disease with GIS. GIS technology powers solutions for telecommunications, utilities, agriculture, defense, oil, health care, transportation, mining, environmental management, petroleum, water/waste-water, and many other industries as well as local, state, and federal government agen-



cies.

Desktop GIS represents the real world on a computer similar to the way maps represent the world on paper. Both GIS and paper maps convey information about places. However, desktop GIS has power and flexibility that paper maps lack. The scale of the map influences the size of what appears on it. With GIS, however, you can store and link huge amounts of information about the objects represented on maps. These objects are called features. Each map feature has a location, a representative shape, and a symbol that represents one or more of its characteristics. Because features on maps are organized according to relative location or position, maps are good for showing the relationships among feature locations. These relationships, called spatial relationships, are important because understand-



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ing them helps us solve problems.

Maps use three basic shapes– points, lines and areas to represent real-world objects. Points represent objects that have discrete locations and are too small to be depicted as areas. Lines represent objects that have length but are too narrow to be depicted as areas. Areas represent objects too large to be depicted as points or lines. Shapes alone do not give you enough information, so maps use graphic symbols to help identify features and provide information about them. Most features can be represented as more than one shape. The scale of a map tells how the size of the map features compares with the size of the geographic objects they represent. Map scales vary from small-scale to large-scale.

With desktop GIS, you are not limited to the amount of



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information you can get about what you see on the map. Desktop GIS stores all the information about map features in a GIS database and links the features on the map to the information about them. This means that you can access all the information about a feature by simply clicking on it.

The information that a desktop GIS stores about map features is referred to as attribute information, or attributes. The attributes of a river, for example, might include its name, length, average depth, rate of flow, water quality, how many dams are on it, and how many bridges cross it. Desktop GIS formats attributes in rows and columns, and stores them as tables. Each column stores a different attribute and each row relates to a single feature. The link between map features and their attributes is the basic principle behind how a desktop GIS works, and is





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the source of its power. Once the map features and attributes are linked, you can access the attributes for any map feature or locate any feature from its attributes in a table. Desktop GIS links sets of features and their attributes and manages them together in units called themes. A theme contains a set of related features, such as roads, streams, parcels, or wildlife habitat areas, along with the attributes for those features.

2.3. Geobrowser

A Geobrowser is better explained with reference to an internet browser, i.e. Internet Explorer. In short, a geobrowser can be understood as an Internet Explorer for geographic information. Like the internet it allows the combination of many types of geographic data from many different sources. The biggest difference





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between the World Wide Web and the geographic web however is that everything within the latter is spatially referenced. Google Earth is the most popular geobrowser available.

A Geobrowser is a client in a web browser or a stand-alone application that is capable of accessing georeferenced data and providing multi-dimensional visualization to end users. It aims as improving a user's experience by providing better maps, faster services and high resolution data. One of the strengths is its ability to provide content and processing power for other web applications. It has the following capabilities:

- It has a robust search capability
- It has real time monitoring and access to historical data
- It provides simple access to data collected by different scientific disciplines through the internet



- It has built-in tools for importing and exporting data

2.4. Web-based GIS

Web GIS is a mechanism to deliver maps and GIS services on the internet. Web-based GIS, or WebGIS, are online GIS applications are excellent data visualization tools. Their functionality is limited compared to software stored on the computer, but they are user-friendly and particularly useful as they not required data download. It is a viable tool for building various geospatial Web application. It uses a centralized paradigm approach. It is a cost effective, easy and efficient way to access and disseminate the geographic information, spatial analysis tools and geospatial web services. It evolves from different web maps and can be modeled as client - server architecture or thin client architec-



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ture. Web GIS is similar to the client/server typical three-tier architecture.

2.4.1. Thin Client Server Approach

1. Advantages of thin client server approach:
 - Central Control
 - Easy for data version
 - Generally cheaper
 - Integration possibilities
 - Regarding some cartographic aspects such font
2. Disadvantages of thin client server approach
 - Large data volume,
 - Slow response time,
 - Less interactive, and
 - No local accountability



2.4.2. Thick Client Server Approach

1. advantages:

- Document/graphics standards are not required
- Vector data can be used
- Image quality not restricted to GIF and JPEG
- Modern interface is possible ; it is not restricted to single – click operations .

2. Disadvantages

- Additional software requirement, and b) Platform/ browser incompatibility

2.4.3. Web GIS Architectures

The Web-GIS prototype implements two parts:



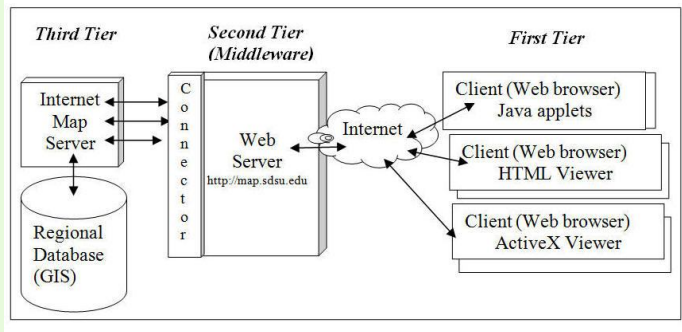
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1. A (server-side) Web Feature Service (WFS) which interacts with clients and processes queries, retrieves and stores data to DBMS and generates WFS/OGC-conformal query responses.
2. A client, implemented as a JavaScript Class, is provided to query data, generate WFS requests and visualise geo-data. The client implements WFS and WMS interfaces and can therefore be used independently from the prototype's server-side implementation.

The geo-processing breaks down into server-side and client-side tasks. A client typically is a Web browser. The server-side consists of a Web Server, Web GIS software and Database .



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Revision Questions

EXERCISE 1.  Describe the different software used in GIS.

Example . Distinguish the different types of client servers.

Solution: Revise.

EXERCISE 2.  What are the characteristics of Geobrowser?



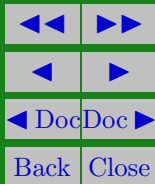
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Solutions to Exercises

Exercise 1. Revise.

Exercise 1





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Exercise 2. Revise.

Exercise 2

